

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

EC-6005(3)-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Statistical Signal Processing

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) What is adaptive filtering? Explain types of adaptive filters.
b) Explain how least mean square filter works.
2. a) Explain the Keiser window technique for designing FIR filter.
b) Explain about estimation Wiener filter.
3. Explain in detail about the Kalman Filtering.
4. a) Explain detection of signals in additive white Gaussian noise using adaptive filter.
b) Explain how extended kalman filter is used to predict and estimate non-linear signals.
5. a) Write down the applications of KF and EKF in audio and speech signals detection. Why is Kalman Filtering so popular?
b) Explain Cholesky Decomposition method with the help of a suitable example.

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6.
 - a) Differentiate the FIR Wiener filter and IIR Wiener filter.
 - b) Explain
 - i) Causal linear Prediction
 - ii) Wiener Deconvolution
7. Which of the following methods are instances of optimal filtering and which are adaptive filtering:
 - i) Wiener Filtering
 - ii) LMS
 - iii) NLMS
 - iv) RLS
 - v) LSJustify your answers.
8. Write short notes on (any two):
 - i) State Estimation Filters
 - ii) Wavelet Transform
 - iii) Inverse modeling
